



Towards Steering without Sacrifice: Principled Training of Steering Vectors for Prompt-only Interventions

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Background: Steering Vectors (SVs)

- What are steering vectors (SVs)?
Add a fixed, scaled vector to hidden states at inference time:

$$\Phi(\mathbf{h}; \alpha, \mathbf{v}) = \mathbf{h} + \alpha \mathbf{v}.$$

Φ : intervention notation, α : steering factor, $\mathbf{v}$: steering direction.

–Lightweight, interpretable alternative to fine-tuning and more robust than prompting.

- Goal: Concept-based steering.
To incorporate a concept (e.g., “Golden Gate Bridge” concept) into response using SV intervention while fulfilling instruction.

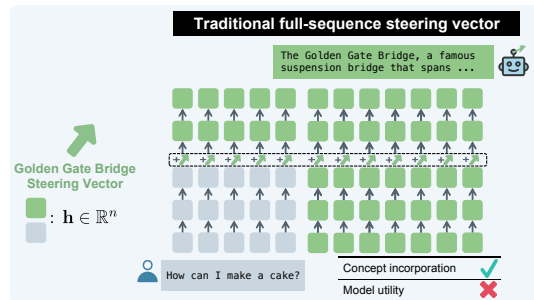


Figure: Traditional full-sequence SV (FSSV). The output incorporates target concept but overlooks input instruction.

Problem

- Two unsolved problems with fine-tuned SVs.
 - Problem 1: Post-hoc factor selection.**
Brittle, requires per-SV grid search at inference time
 - Problem 2: Degraded model capabilities.**
Full-sequence intervention degrades generation quality even with careful factor selection.

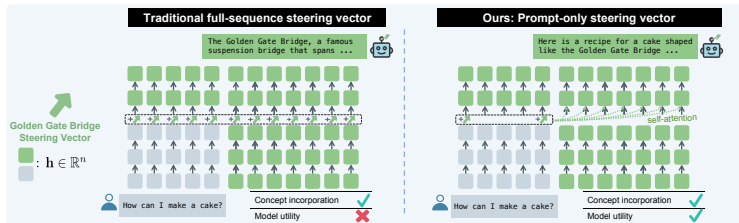


Figure: FSSV vs. PrOSV.

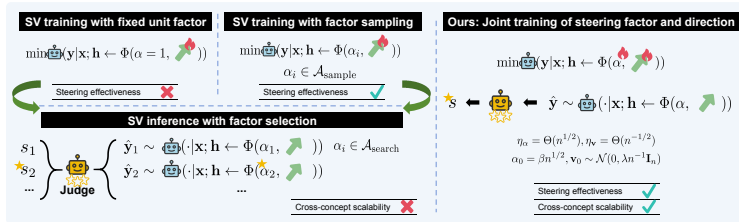


Figure: Factor selection vs. end-to-end training.

Method

Research questions

- Can we train steering factors in an end-to-end manner, such that factor selection is not required?
- Can we incorporate target concept while maximally preserving model capabilities?

Method: Part 1 – Joint Training based on Scaling Theory Analysis

- **Key insight:** Treat α as a trainable parameter, not a constant.
Setting $\alpha = 1$ does not work.
- **Challenge:** How do we set learning rates and initialization for α and $\mathbf{v}$ jointly?
- **Solution:** Neural network scaling theory (maximal parameterization, μP framework).
 - Analyze training dynamics in the infinite-width limit.
 - Derive that moderately large α_0 and learning rate η_α are essential for *stable, efficient* joint learning.
 - Too small $\rightarrow \alpha$ is trapped locally; too large $\rightarrow$ instability.
- **Result:** One-time hyperparameter tuning before training.
No per-SV factor selection at inference time.

Algorithm 1 Our scheme for joint training of SV factors and directions.

input Training set $\mathcal{D}$, direction learning rate $\eta_v = \Theta(-\frac{1}{2})$, factor learning rate $\eta_\alpha = \Theta(\frac{1}{2})$, direction initialization size λ , factor initialization size β , training steps T , loss function $\ell(\cdot)$

output Steering factor α_T , steering direction $\mathbf{v}_T$
 $\mathbf{v}_0 \sim \mathcal{N}(\mathbf{0}, \lambda n^{-1} \mathbf{I}_n)$ {To ensure: $\mathbf{v}_0 = \Theta(-\frac{1}{2})$ }
 $\alpha_0 \leftarrow \beta n^{1/2}$ {To ensure: $\alpha_0 = \Theta(\frac{1}{2})$ }
 $t \leftarrow 0$

while $t < T$ **do**

$(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}$

$l_t \leftarrow \ell(p_\Phi(\cdot | \mathbf{x}; \mathbf{h} \leftarrow \Phi(\mathbf{h}; \alpha_t, \mathbf{v}_t)), \mathbf{y})$

$\{g_t^v, g_t^\alpha\} \leftarrow \text{Adam}(\{\nabla_v l_t, \nabla_\alpha l_t\})$

$\mathbf{v}_{t+1} \leftarrow \mathbf{v}_t - \eta_v g_t^v$

$\alpha_{t+1} \leftarrow \alpha_t - \eta_\alpha g_t^\alpha$

$t \leftarrow t + 1$

end while

Method: Part 2 – Prompt-Only Steering Vector (PrOSV)

- **Inspiration:** Representation fine-tuning (ReFT) shows prompt-only interventions suffice for task adaption.
- **Design choices explored:** prompt prefix and/or suffix.
- **Why does it work?**
Mechanism: Decoded tokens attend to intervened prefixes/suffixes without damaging global attention patterns.

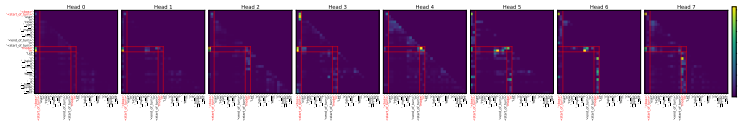


Figure: |PrOSV - Original| attention weights.

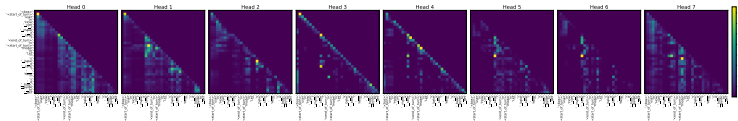


Figure: |FSSV - Original| attention weights.

Experiment: PrOSV Intervention Location Sweep

- Models: Gemma2-2B-IT (G2B), Gemma2-9B-IT (G9B), Qwen2.5-32B-IT (Q32B).
- Layers: middle layers–10th of G2B, 20th of G9B, 32nd of Q32B.
- Hyperparameters:
 - η_α : 1e-2, 1e-1, 1, 10, 20, 40.
 - α_0 : 1, 2, 4, 8, 16.
- Baselines:
 - FSSV: Traditional full-sequence SV.
 - Full prompt: Intervene on every prompt token but no decoded tokens.
 - p_k : k prompt prefix tokens.
 - s_k : k prompt suffix tokens.
 - p_j+s_k : j prefix tokens and k suffix tokens.
- Metrics: Highest grid-selected concept/overall scores.

Table: Highest grid-selected overall (O) and concept (C) scores on Concept10. Best results are highlighted in bold.

Location	G2B; L10		G9B; L20		Q32B; L32	
	O	C	O	C	O	C
FSSV	0.65	0.97	0.86	1.17	0.93	1.27
Full prompt	0.54	1.12	0.71	1.41	0.88	1.58
Budget: 2 tokens						
p_2	0.65	0.69	0.78	0.90	0.91	1.00
s_2	0.68	0.82	0.71	0.94	1.11	1.28
p_1+s_1	0.67	0.79	0.91	1.12	1.10	1.24
Budget: 4 tokens						
p_4	0.59	0.63	0.73	0.85	0.94	0.97
s_4	0.69	0.83	0.77	1.03	1.08	1.24
p_2+s_2	0.70	0.82	0.92	1.14	1.16	1.33
Budget: 8 tokens						
p_8	0.58	0.64	0.75	0.85	0.90	0.94
s_8	0.61	0.85	0.74	1.12	0.92	1.24
p_4+s_4	0.69	0.85	0.89	1.09	1.13	1.30

Experiment: Large-Scale Concept-based Steering on AxBench

- Benchmark: AxBench (500 concepts).
- Metric: Overall steering score.
- Methods:
 - Prompt: Steering prompt (e.g., “please mention Golden Gate Bridge”)
 - FSSV baseline: only train steering direction, post-hoc factor selection.
 - Lang.: minimize cross-entropy loss.
 - SimPO: SimPO preference optimization objective; use contrastive data.
- Result:
 - Naive joint training with $\alpha = 1$ yields useless SVs.
 - ProSV achieves SOTA with SimPO objective.

Table: Overall steering scores (0–2; $\uparrow$) on AxBench. Best results are highlighted in bold.

Method	$\mathcal{D}_{L10}^{G2B}$	$\mathcal{D}_{L20}^{G9B}$	$\mathcal{D}_{L32}^{Q32B}$
Prompt	0.698*	1.075*	1.060
Objective: Lang.			
FSSV	0.663 $\uparrow$	0.788 $\uparrow$	0.798
+ Fixed unit factor*	0.072	0.024	—
+ Joint training	0.736	0.821	0.919
ProSV	0.758	0.859	1.049
Objective: SimPO			
FSSV (RePS)	0.756 $\uparrow$	0.892 $\uparrow$	0.947
+ Joint training	0.769	0.886	0.982
ProSV	0.803	0.905	1.102
LoReFT*	0.701	0.777	—
DiffMean*	0.297	0.322	—
SAE ‡	—	0.546	—

Experiment: Model Utility & Robustness

- Does steering hurt the model? Tested on tinyMMLU and tinyGSM8K.
- Metrics:
 - Overall steering score under concept suppression attack (e.g., “do not mention the Golden Gate Bridge”).
 - Accuracy on tinyMMLU/tinyGSM8K.
- Results: PrOSV sits on a better Pareto frontier than FSSV.

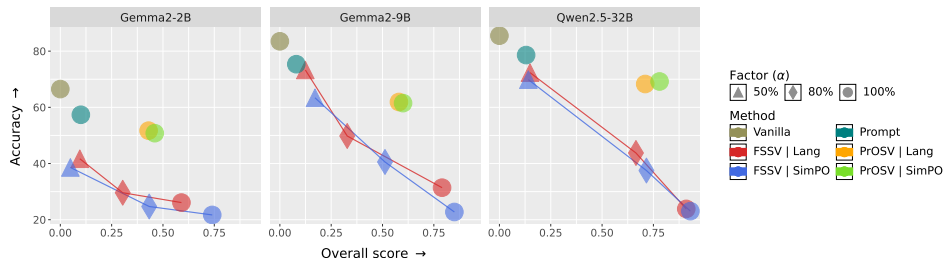


Figure: Average accuracy on tinyMMLU and tinyGSM8K vs. overall steering scores under concept suppression attack. We adjust FSSV steering factor to form a performance frontier.

Takeaway

- Two principled departures from conventional SV practice:
 - Joint training of α and $\mathbf{v}$ —grounded in scaling theory $\rightarrow$ eliminates inference-time factor selection
 - Prompt-only intervention —4 tokens is enough $\rightarrow$ preserves model utility and robustness
- Limitations & Future Work:
 - Focus on fine-tuned SVs; optimization-free SVs not covered.
 - More general token selection strategies beyond prefix/suffix.
 - Fundamental tension between robustness and utility not fully resolved.